Earnings Call Transcript Analysis: Predicting Stock Price Movements

Author: Matthew Hawkes

Date: 4/27/26

Objective

Guiding Question: Do executives' words during earnings calls contain predictable signals about short-term stock price direction?

The objective of this project is to build a machine learning pipeline that utilizes NLP-derived features from transcript text to predict both the direction and magnitude of stock movement after earnings calls.

Executive Summary

I developed a thorough machine learning pipeline that used Natural Language Processing (NLP) to analyze earnings call transcripts to determine both the direction and magnitude of short-term swings in a stock.

The workflow consisted of:

1. Dataset Preparation
2. Exploratory Data Analysis
3. Preprocessing & Feature Engineering
4. Model Comparison & Selection
5. Tuning Top-Performing Models
6. Final Predictions & Evaluation

Understanding the Data

To build the ML models, two datasets were needed: the earnings calls transcripts and the actual stock movement.

Dataset 1 — Motley Fool Transcripts:

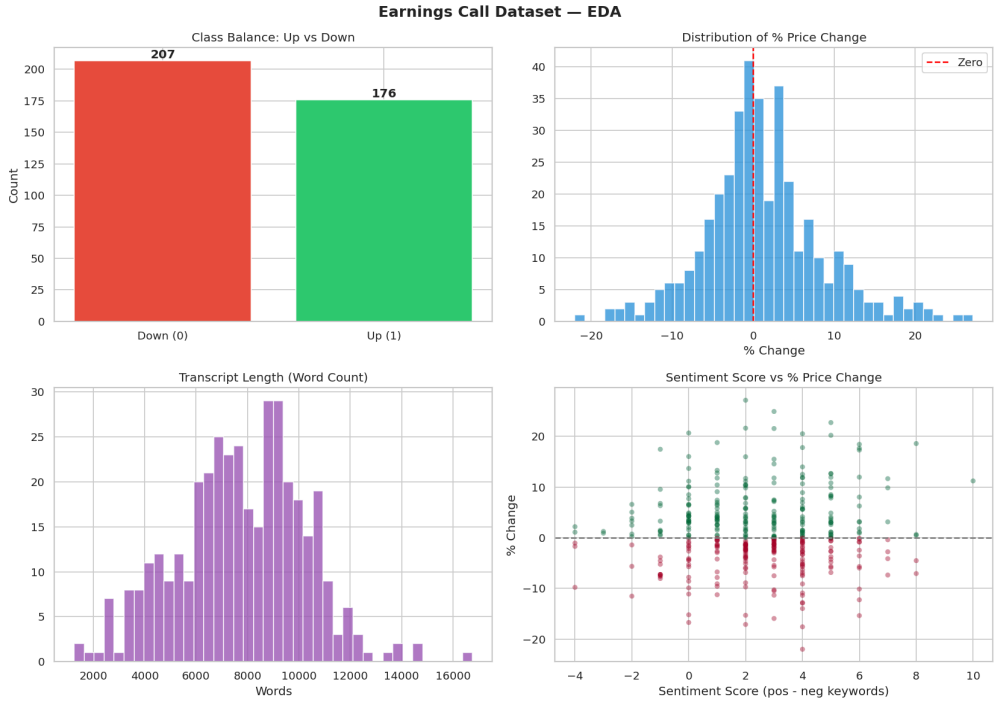
- Scraped quarterly earnings calls from Motley Fool
- Fields: ticker symbol, earnings date, full transcript text
- Covers multiple quarters across large-cap US stocks
- 500 transcripts sampled for this project
- Transcripts average ~8,000–10,000 words each

Dataset 2 — Yahoo Finance Data

- Yahoo Finance historical OHLCV data via yfinance API
- Price before: closing price on or before earnings date
- Price after: closing price 3 trading days post-earnings
- Binary target: 1 = price up, 0 = price down
- Continuous target: % change between before and after

I combined these two data sources to build a singular dataset the models could understand and learn from.

Exploratory Data Analysis (EDA)



1. **Class Balance:**
    - There was a mild class imbalance between stocks trending up or down.
    - It was not significant enough to require stratification, preserving class distribution, in any step.
2. **% Price Change:**
    - The price change was generally normally distributed, showing no bias towards positive or negative shifts in stock prices.
3. **Transcript Length:**
    - Transcripts were on average 8,500 words in length.
    - There were no significant patterns in length that could lead to discernible signal
4. **Sentiment Score vs % Price Change:**
    - This graph appeared to be left-shifted, as there was a larger concentration of data points with a positive sentiment score.
    - This matches the intuition that executives tend to frame things more positively than negatively.
    - Interestingly, more neutral transcripts led to greater price changes, while more negative or positive sentiment scores led to smaller price changes.

Preprocessing & Feature Engineering

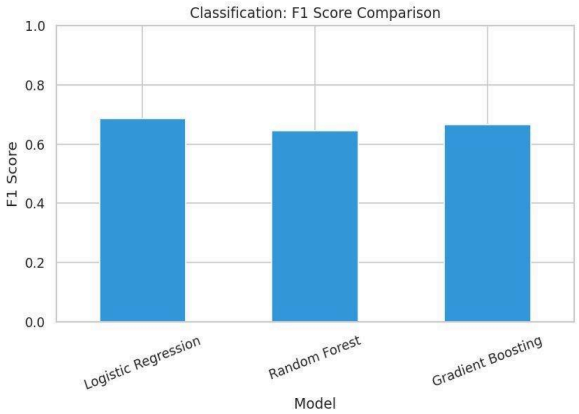
I cleaned the text by lowercasing it, stripping URLs, brackets, and punctuation, collapsing whitespace, and removing transcripts of fewer than 50 words.

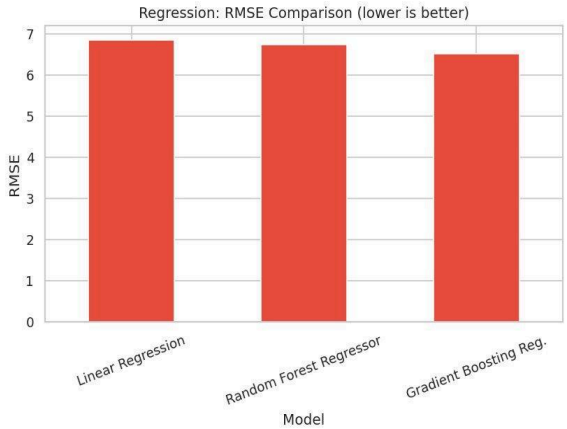
Next, I vectorized the text using TF-IDF, yielding a sparse matrix with 5,000 unigram and bigram features.

Finally, I created 4 structured features: transcript word count, positive keyword count, negative keyword count, and most importantly, a net sentiment score, which helped determine whether a company was performing well or not.

I combined all these features to build a matrix that each model can use.

Baseline Model Comparison





Out of the classification models, no model performed significantly better than the rest. However, taking into account the context of the problem, it seemed more logical to capture the nonlinear relationships of human speech with the random forest model.

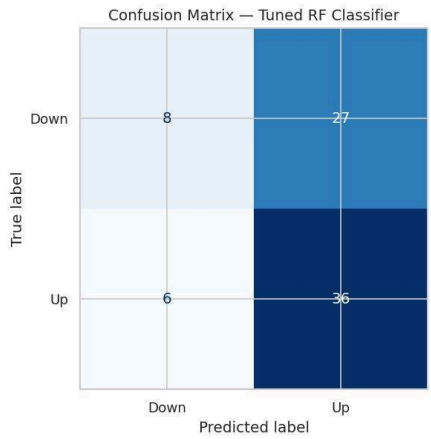
For the regression models, each model similarly performed practically equally; gradient boosting did slightly better, so I went ahead with this one.

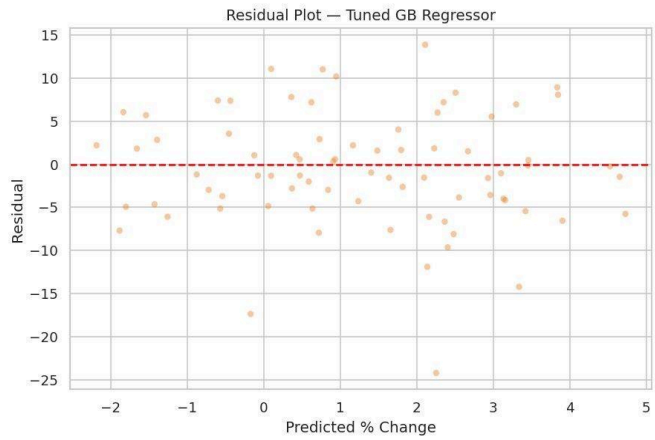
Hyperparameter Tuning

Tuning the gradient boosting regression model was scored on neg-RMSE, and the selected variables were n_estimators, learning_rate, max_depth, and subsample.

Tuning the random forest classifier was scored on F1, and the parameter grid included n_estimators, max_depth, min_samples_split, and max_features.

Final Model Evaluation





Classification Summary:

- **Confusion Matrix:**

  RF Classifier predicts UP heavily (36 correct, 6 false negatives). Strong recall for up-moves; weaker at catching real down-moves. Reflects mild class imbalance.

- **Accuracy:** 57%
- **Precision:** 58%
- **Recall:** 86%
- **F1 Score:** 0.79
- The model did not excel but performed better than just randomly guessing.

Regression Summary:

- **Residual Plot:**

  Residuals scatter randomly around zero, indicating no systematic bias. Spread reflects the inherent unpredictability of short-term price changes from text alone.

- **RMSE:** 6.6%
- **MAE:** 4.5%
- **R^2:** 0.04
- These results show that the model can be highly inaccurate, and using this to determine the magnitude of stock swings is not reliable.

Limitations & Challenges

Weak Signal from Text

Earnings calls are carefully scripted. Executives rarely reveal material information that hasn't already been priced in, creating a fundamental gap between language and price movement.

Market Randomness

Short-term stock movements are driven by macro factors, sector momentum, and market microstructure, which are forces entirely outside the transcript. This explains the low regression R^2.

Data Merging Gaps

yfinance returns no data for tickers on trading holidays, suspended stocks, or delisted companies, reducing the usable dataset and potentially introducing survivorship bias.

No Temporal Split

A random train-test split may allow look-ahead bias. A chronological split (train on older calls, test on newer) is more realistic but reduces training data size for smaller datasets.

Future Improvements

FinBERT Embeddings

Replace TF-IDF with domain-specific pre-trained transformer embeddings (FinBERT) trained on financial text for richer semantic representations.

Q&A vs. Remarks

Separate the prepared executive remarks from the analyst Q&A section. The Q&A may contain more genuine signals as executives respond to probing questions.

Macro Features

Add sector, market cap, analyst consensus estimate vs. actual EPS/revenue as structured features to give models a richer context beyond text alone.

Conclusion

Classification

Random Forest F1 ~0.69. The transcript text does carry a directional signal, though it is modest.

Regression

RMSE ~6.6% with near-zero R^2 confirms that predicting the magnitude of price moves from text alone is unreliable. To do so, structural market data is needed.

Key Takeaway

Earnings calls provide weak but real NLP signals for direction. This project demonstrates a complete, reproducible pipeline to create sophisticated models.